University College of Engineering, Tindivanam

(A Constituent College of Anna University, Chennai)

PROJECT TITLE: Deep Learning–Based Tamil Handwritten Character Recognition System

TEAM MEMBERS

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LITERATURE SURVEY

S.No.	Paper (Author, Title, Journal Name, Year)	Key Idea / Approach	Advantages	Disadvantages	Evaluation Metrics	Data Set (if required)
1.	Jayachandran, S., Selvakumar, Y., & Pearline, S. A., **"Multi-Attention Based Convolutional Neural Network for Tamil Handwritten Character Recognition"**, **IEEE Access**, 2025	Uses **EfficientNet-B0 with SE and CBAM attention** to recognize handwritten Tamil vowels by focusing on important features and character regions. **Models Used:** 1. CNN 2. EfficientNet-B0 3. Custom CNN architecture **Activation Function Used:** 1. ReLU – hidden layers 2. Softmax – output layer **Feature Extraction Model:** 1. CNN Convolution Layers 2. EfficientNet-B0 (pretrained) **Pooling Method:** Max Pooling **Output:** Single handwritten Tamil character prediction	1. ~92% – 97% Accuracy rate. 2.Automatic feature extraction 3.Handles handwriting variations 4.Suitable for image-based recognition	1.Needs large dataset 2.Confusion between similar characters 3.High training cost	1. Accuracy 2. Recall 3. Precision 4. F1 Score 5. ROC-AUC	1.**HP Labs Tamil Dataset** – HP Research Lab 2.**HTV-13 Dataset** – IEEE / Research source 3.**Tamil Handwritten Dataset** – Kaggle

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2.	Vinotheni & Lakshmana Pandian, **"*End-To-End Deep-Learning-Based Tamil Handwritten Document Recognition and Classification Model*", IEEE Access**, 2024	Proposes an **end-to-end deep learning model** for Tamil handwritten documents using **median filtering**, **line & character segmentation**, **MobileNet for feature extraction**, and **BiGRU with Water Strider Optimization** for recognition. **Models Used:** 1. MobileNet + BiGRU 2. CNN (MobileNet backbone) 3. WSO (optimization) **Activation Function Used:** 1. ReLU – CNN layers 2. Sigmoid – GRU gates 3. Softmax – output layer **Feature Extraction Model:** MobileNet (Depthwise Separable CNN) **Pooling Method:** Max Pooling **Output:** Recognized Tamil handwritten text (line/character level)	1 98.48% (maximum average accuracy) 2. End-to-end recognition 3. Low computation (MobileNet) 4. Real-time document processing	1. Complex architecture 2. Needs segmentation accuracy 3. Training time high 4. Dataset not publicly large	1. Accuracy 2. F1 Score 3. Recall (Sensitivity) 4. Specificity 5. Precision 6. AUC	Self-collected Tamil handwritten document dataset

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3.	Periyasamy, H., Natarajan, S., Murugappan, M., & Chowdhury, M. E. H., "**ResTANet: A Deep Residual Neural Architecture for Tamil Handwritten Character Recognition**", **IEEE Access**, 2025	Introduces **ResTANet**, a **deep residual CNN architecture** designed for Tamil handwritten character recognition. Uses **residual connections** to learn deep features effectively and reduce vanishing gradient problems. **Models Used:** • ResTANet • Deep Residual CNN • Attention-based CNN **Activation Function Used:** • ReLU – convolution layers • Softmax – output layer **Feature Extraction Model:** Residual CNN blocks (ResNet-style) **Pooling Method:** Max Pooling **Output:** Single handwritten Tamil character	1. 98.12% accuracy 2. Deep residual learning 3. Reduces vanishing gradient 4. Handles complex Tamil characters	1. High computational cost 2. Large training time 3. Requires large dataset 4. Complex architecture	1. Accuracy 2. Precision 3. Recall 4. F1-Score	**HP Labs Tamil Handwritten Dataset-** HP Research / IEEE

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4.	Jayanthi, V., & Thenmalar, S., ***"Tamil OCR Conversion from Digital Writing Pad Recognition Accuracy Improves through Modified Deep Learning Architectures"***, Journal of Sensors, 2023	Proposes a **Tamil OCR system using digital writing pad input**, where handwritten strokes are captured directly and recognized using **modified deep learning architectures** to improve recognition accuracy **Models Used:** 1. CNN 2. LSTM 3. Hybrid CNN–LSTM **Activation Function Used:** 1. ReLU – CNN layers 2. Tanh / Sigmoid – LSTM 3. Softmax – output layer **Feature Extraction Model:** CNN (Spatial feature extraction) **Pooling Method:** Max Pooling **Output:** Recognized Tamil characters from digital writing pad	1. 99.1% Accuracy 2. Uses digital writing pad data, reducing noise compared to scanned images. 3. Works in real time. 4. Reduces manual digitization	1. Requires digital writing pad 2. Not suitable for offline images 3. Hardware dependency 4. Higher system cost	1. Accuracy 2. Precision 3. Recall 4. Recognition Rate	Self-collected online Tamil handwriting dataset

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5.	Manoj K, Iyapparaja M, ***"Tamil Handwritten Character Recognition: A Comprehensive Review of Recent Innovations and Progress",*** International Journal of Advanced Computer Science and Applications,2025	Proposes a Tamil handwritten character recognition system using image preprocessing, feature extraction, and machine learning / neural network–based classification to recognize individual Tamil characters from handwritten inputs. **Models Used:** 1. CNN 2. Artificial Neural Network (ANN) 3. Hybrid CNN–ANN **Activation Function Used:** 1. ReLU – hidden layers 2. Softmax – output layer **Feature Extraction Model:** CNN convolution layers **Pooling Method:** Max Pooling **Output:** Recognized handwritten Tamil character	1. ~94% – 96% Accuracy . 2. Automatic feature extraction 3. Suitable for offline handwriting	1. Confusion in similar characters . 2. Focuses only on character-level recognition. 3. Dataset size is limited.	1. Accuracy 2. Precision 3. Recall 4. Recognition Rate	**Tamil Handwritten Character Dataset**- kaggle